Kalman and Levinson Filtering , 3 credits

1 Course information:

This course presents an introduction to Kalman and Levinson filtering with applications to feedback control. By the use of the Projection Theorem and Hilbert Spaces, students build the intuition of how the Kalman Filter is developed from linear spaces, with the inclusion of Random Variables. From there, students are presented with different ways on how the Discrete-time Kalman Filter is derived and other ways the Kalman Filter can be written to suit different cases. Students are also briefly shown the Continuous-time Kalman Filter and are taught how to apply both the Discrete-time and Continuous-time Kalman Filters in Feedback Systems. The students will be introduced to the Extended Kalman Filter and the Unscented Kalman Filter and are expected to be able to apply them in certain nonlinear estimation problems. Finally, the students will learn about Wide-Sense Stationary Processes and deeper Fourier Analysis to set up the understanding for the Levinson Filter.

2 Course Description

1. The Projection Theorem
2. Least squares and random vectors
3. The discrete time Kalman filter
4. Kalman prediction and smoothing
5. The information Kalman filter
6. The steady state Kalman filter
7. The continuous time Kalman filter
8. Kalman filtering and state feedback
9. Matlab and Simulink examples
10. Extended Kalman Filter
11. Unscented Kalman Filter
12. Wide sense stationary processes
13. The spectral density
14. The Wiener filter
15. The Levinson filter
16. The Gohberg-Semencul Toeplitz inversion formula

3 Learning Resources and Reference Text

Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook: The textbook is a set of notes. The notes are posted on Brightspace in a .pdf file free of charge.

4 Assignments

The course assigns homework which is graded and included in part of the final grade. There may also be one or two special projects in the course. The homework will be handed in by gradescope or brightspace. The solutions for the homework will be provided. No late homework will be accepted unless there is a medical or university related reason.

5 Course Learning Objectives

On completing this course the student shall be able to:

1. A theoretical and practical prospective of Linear Estimation Theory.
2. A thorough understanding of the derivation and the construction of the Kalman Filter.
3. Applying the Kalman Filter in real-world applications.
4. The ability to apply the Extended Kalman Filter and the Unscented Kalman Filter in certain nonlinear filtering estimation problems.
5. An understanding of Wide Sense Stationary Processes and Levinson Filtering.

6 Grading scale

- Grades are A, B, C, D, F and no $\pm$ grades.
- The homework counts for 35% of the final grade. The exams, final exam and projects will account for 65% of the final grade.
- The final grade is determined by a curve. The instructor sorts all the grades and then calculates the mean, median and standard deviation. Based on this data an A is a minimum of 35% of the class (and could be a higher percentage). The rest of the grades are B with the exception of usually a very few students which did very little for the course. They may receive a C, D or F grade.

- The grading and distribution of grades in this course are done totally blind. The instructor has no idea what student is associated with any score when assigning grades.

7 Artificial Intelligence

Students are not allowed to use any Artificial Intelligence or Large Language Models (AI/LLM) to generate parts of a homework or exam solutions. Use of AI or LLMs will be considered a blatant attempt at cheating and will result in a failing grade in the course.